Food Remedy API

Historical Timeline of Food Additive Usage and Regulation – RE038

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Date: 21 August 2025

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Introduction

The modern food industry relies heavily on additives to maintain food safety, prolong shelf life, and improve sensory qualities such as taste, colour, and texture. In the European Union and other regions, these additives are regulated through the E-number system, which designates substances that have undergone rigorous safety evaluations. However, the presence of an E-number does not guarantee universal safety, as some additives may still pose risks to sensitive individuals. This report proposes the creation of a structured, multi-tiered taxonomy that links each E-number to its chemical category, technological function, and associated health risk. A key feature of this model is its ability to distinguish between true allergic reactions and those involving IgE-mediated immune responses, and other types of adverse effects, such as hypersensitivity or pharmacological responses. This distinction is essential to ensure accurate communication of health risks to consumers without causing unnecessary alarm.

The report is structured to deliver a comprehensive analysis. It begins by examining the regulatory frameworks of key international authorities, notably the European Food Safety Authority (EFSA) and Food Standards Australia New Zealand (FSANZ), to reveal differing global approaches to food additive assessment. It then introduces a formal backend taxonomy, supported by a detailed technical data model for efficient classification and tagging. The central analysis explores high-risk additive groups, assessing their roles in food production, reported health impacts, and supporting clinical evidence. The report concludes with practical recommendations for maintaining data integrity and system reliability, ensuring that the proposed structure remains a sustainable and trustworthy reference in the evolving landscape of food regulation.

European Union Framework: EFSA Mandate

The European Food Safety Authority (EFSA) plays a vital role in regulating food additives within the European Union. It is responsible for providing scientific guidance to the European Commission, which works alongside Member States to determine the approval and usage conditions of such substances. EFSA conducts a structured four-step risk assessment to ensure the safety of food additives before they are authorised for use.
Which Involves:

1. **Hazard identification** involves recognising the potential negative health effects that a particular substance may cause.
2. **Hazard characterisation** focuses on evaluating the relationship between the amount consumed and its impact on health, usually resulting in a safety benchmark such as the Acceptable Daily Intake (ADI), which refers to the quantity that can be safely consumed every day over a lifetime.
3. **Exposure assessment** estimates how much of the additive an individual is likely to consume through their regular diet.
4. **Risk characterisation** then compares this estimated intake with the ADI to determine whether the additive's use is considered safe.

Australia and New Zealand (ANZ) Framework

Food Standards Australia New Zealand (FSANZ) serves as the main regulatory body for food additives in its region, operating under the internationally recognised Codex Alimentarius framework. Like the European Food Safety Authority (EFSA), FSANZ conducts both hazard and dietary exposure assessments to determine an Acceptable Daily Intake (ADI), supported by extensive testing, including animal studies, to establish safe concentration levels. Approval is granted only if the expected intake remains within the safety threshold. FSANZ also engages in ongoing monitoring of scientific research and collaborates with international organisations such as EFSA and the US FDA to revise regulations as new evidence emerges. While this highlights a global effort to ensure food safety, regional regulatory differences persist.

Comparative Table Between EFSA vs FSANZ

Additive Name	*E-Number*	*Regulatory Status in EU*	*Regulatory Status in FSANZ*	*Explanation of Discrepancy*
Amaranth	E123	✅	✅	Banned in the US in 1977. Later studies by EFSA and JECFA concluded it is not carcinogenic.
Vegetable Carbon	E153	✅	✅	Not permitted in the US due to concerns about the presence of polycyclic aromatic hydrocarbons in its production. EFSA's review came to a different conclusion.
Butylated Hydroxyanisole (BHA)	E320	✅	✅	Authorized in both the EU and the US for a wide range of foods. This illustrates a case of regulatory alignment.
Ammonium Phosphates	E342	❌	✅	Permitted in Australia and New Zealand. Not listed as approved in the EU, possibly because no application was ever submitted.

Hierarchical Structure

- **Level 1 – E-Number:** This refers to the specific code assigned to a food additive, such as E220, which helps identify the substance within regulatory systems.
- **Level 2 – Chemical Group:** This level categorises the additive based on its chemical family. The E-number itself often hints at the group it belongs to, as certain numerical ranges correspond to specific types of compounds.
- **Level 3 – Functional Role:** At this stage, the additive is classified according to its main function in food processing, such as acting as a preservative, colourant, or flavour enhancer. It is important to recognise that many additives have more than one role, so the system should allow for multiple classifications.
- **Level 4 – Allergen and Health Risk Classification:** This final level addresses health-related information by identifying potential adverse effects. It differentiates between confirmed allergies, sensitivities, behavioural impacts, and other health risks, making it the most vital layer for consumer safety awareness.

Key Additive E-Number Ranges and Function Table:

E-Number Range	*Primary Function*
E100–E199	Colours
E200–E299	Preservatives and acids
E300–E399	Antioxidants and acidity regulators
E400–E499	Thickeners, stabilisers, and emulsifiers
E500–E599	Acidity regulators and anti-caking agents
E600–E699	Flavour enhancers
E700–E799	Antibiotics
E900–E999	Glazing agents, gases, and sweeteners
E1000–E1599	Additional additives

Additive Groups

1.0 Preservatives are added to food to prevent spoilage caused by microorganisms and oxidation, helping to maintain safety and extend shelf life. However, some types of preservatives have been widely reported to cause negative health reactions in certain individuals.

1.1 The Sulphite Group (E220-E228)

The sulphite group is a family of preservatives that includes:

- sulphur dioxide (E220)
- sodium sulphite (E221)
- sodium hydrogen sulphite (E222)
- sodium metabisulfite (E223)
- potassium metabisulfite (E224)
- calcium sulphite (E226)
- calcium hydrogen sulphite (E227)
- potassium hydrogen sulphite (E228)

Their primary function is to prevent microbial growth and browning, acting as both preservatives and color stabilizers. They are widely used in dried fruits, wines, processed meats like sausages, and various beverages.

The health risk associated with sulphites is classified as a high-risk respiratory hypersensitivity. Exposure can induce a range of adverse clinical effects, but the most common and concerning are respiratory symptoms, particularly in asthmatic individuals. It is estimated that sulphite sensitivity affects 3 to 10% of people with asthma, with symptoms including wheezing, chest tightness, coughing, and shortness of breath.

1.2 The Benzoate & Paraben Groups (E210-E219)

The benzoate group includes:

- benzoic acid (E210)
- sodium benzoate (E211)
- potassium benzoate (E212)
- calcium benzoate (E213).

These substances are used as preservatives to inhibit the growth of bacteria, yeasts, and molds, commonly found in fruit products, soft drinks, and pickled foods. Similarly, parabens (E214-E219) are a group of preservatives used to extend the shelf life of food and other products

Adverse reactions to benzoates and parabens are most commonly linked to skin and respiratory issues. Documented symptoms include skin reactions such as hives (urticaria), redness, itching, and contact dermatitis.

1.3 Colors (E100-199)

Food Colorings are added to enhance the visual appeal of food and drinks. While some are derived from natural sources, many are synthetic, and some have been associated with a range of health concerns.

Azo Dyes Table:

E-Number	*Name*	*Color*	*Health Risk Category*	*Health Effects Description*
E102	Tartrazine	Yellow	May Cause Behavioural or Allergic Response	May Negatively influence Children's Behaviour and attention. Asscociated with Hives, Eczema, and Asthma
E104	Quinoline Yellow	Yellow	May Affect Behaviour	May reduce concentration and increase hyperactivity in children.
E110	Sunset Yellow FCF	Orange	Behavioural and Allergic Risk	May impair attention in children. Also linked to asthma, rhinitis, and skin rashes.
E122	Azorubine	Red/Purple	Behavioural and Allergic Risk	May influence behaviour in children and trigger allergic symptomps such as hives, dermatitis, and asthma.
E124	Ponceau 4R	Red	Behavioural and Allergic Risk	May impact children's focus and is asscociated with allergic reaction such as hives and asthma.
E129	Allura Red AC	Red	Behavioural and Allergic Risk	May disrupt children's behaviour. Known to cause asthma, rhinitis, and skin irritation such as hives.

While azo dyes are a primary concern, other colorants, both natural and synthetic, have been associated with adverse reactions. A prime example is Cochineal (E120), a natural red dye obtained from insects. Despite its natural origin, carmine, a component of cochineal, has been documented to cause intermittent bouts of dermatitis, asthma, and anaphylaxis in sensitive individuals.

1.4 Flavour Enhancers (E600-699)

Monosodium Glutamate (MSG) is a widely used flavour enhancer found in many processed foods, yet it has long been surrounded by controversy. It is often associated with "Chinese Restaurant Syndrome," a collection of reported symptoms such as facial numbness, chest tightness, headaches, and burning sensations. These reactions are not classified as true allergies but rather as hypersensitivity or pharmacological responses, since they are not IgE-mediated. Research also suggests that MSG may worsen conditions such as unstable asthma and chronic hives in sensitive individuals.

1.5 Other Additives

Aspartame (E951): This artificial sweetener is commonly used in sugar-free products. Although recognised as safe by regulatory authorities, it has been associated with a variety of neurological and behavioural effects, such as headaches, dizziness, and mood changes. In addition, some individuals report gastrointestinal and skin-related symptoms, indicating sensitivity to the compound.

Lecithin (E322): As an emulsifier, lecithin represents a distinctive case in food safety considerations. It is a naturally derived substance obtained from sources such as eggs, soy, rapeseed, and sunflower seeds. The health concerns linked to lecithin are not due to the additive itself but arise from residual protein traces from its source ingredients. This creates a challenge for data systems, as food labels may not specify the origin of lecithin. Therefore, for an accurate risk assessment, systems must account for the possibility that E322 could introduce cross-contamination risks from major allergens.

Chronological Timeline Table

YEAR(S)	ADDITIVE	EVENT/CONTEXT	HEALTH CONCERN	REGULATORY ACTION
1968–1969	MSG (Monosodium Glutamate) (E621)	Reports of "Chinese Restaurant Syndrome" brought the additive under public scrutiny, authorities launched formal safety assessments rather than prohibition.	Mild hypersensitivity reactions such as headaches and flushing reported in a small proportion of individuals; not classed as true allergy.	EFSA subsequently established a group ADI of 30 mg/kg body weight per day (2017), authorisation maintained worldwide.
1970S (ESP. 1979)	Tartrazine (E102; FD&C Yellow No. 5)	Scientific debate centred on possible behavioural effects in children and hypersensitivity complaints.	Documented reactions include asthma, urticaria, and reduced attention in susceptible groups.	US: mandatory declaration of Yellow No. 5 on labels (1979). EU: warning statement required from 2010 under the "Southampton Six" provisions.
1976	Amaranth / Red (E123)	Animal studies suggested a carcinogenic potential at high experimental doses.	Suspected cancer risk from rodent studies, though not established in humans.	US FDA prohibited Red No. 2 in 1976. EU and FSANZ continue to permit restricted uses within the acceptable daily intake.
1977–2001	Saccharin	Rat studies indicated bladder tumours, which provoked public and scientific debate.	Initially regarded as carcinogenic in animals through a mechanism later judged irrelevant to humans.	US: warning labels introduced in 1977, subsequently removed in 2001 after delisting from carcinogen registers in 2000.
1981	Aspartame (E951)	Approved by FDA in 1981 and extended to carbonated beverages in 1983.	Some reports of headaches: mandatory warning for phenylketonurics due to phenylalanine content, otherwise considered safe at permitted intakes.	IARC classified as "possibly carcinogenic to humans" (Group 2B) in 2023, while JECFA reaffirmed the ADI of 40 mg/kg bw/day.
2005–2007	Sodium Benzoate (E211)	Investigations revealed benzene formation in soft drinks containing both benzoate and vitamin C under certain conditions.	Benzene is a recognised carcinogen, risk linked to specific product formulations rather	Drinks reformulated to prevent benzene formation, regulators confirmed market

			than the preservative alone.	levels posed no safety concern after changes.
2007–2010	"Southampton Six" dyes (E102, E104, E110, E122, E124, E129)	UK study indicated mixtures of six dyes with sodium benzoate might increase hyperactivity in some children.	Reported behavioural effects such as reduced attention span and heightened activity in a subset of children.	EU: since 2010, all foods containing these colours must state: "May have an adverse effect on activity and attention in children."
2010S	MSG (E621)	Continued reassessment of safety following earlier hypersensitivity concerns.	Occasional transient symptoms observed at high single doses, though not classed as allergy.	EFSA's 2017 evaluation confirmed authorisation, setting an ADI of 30 mg/kg bw/day and recommending monitoring of high consumers.
2021–2022	Titanium Dioxide (E171)	EFSA concluded that potential genotoxic effects of ingested particles could not be excluded.	Possible DNA damage hazard identified; no direct evidence of carcinogenicity in humans.	EU: authorisation withdrawn and a full ban implemented in August 2022. Other regions, such as the US, did not follow suit.
2023	Aspartame (E951)	WHO issued a joint assessment, with IARC identifying a hazard and JECFA assessing risk.	"Possibly carcinogenic" classification by IARC based on limited evidence; JECFA found it safe within the existing ADI.	No regulatory changes: EFSA, FDA, and JECFA all maintained current approvals and ADI levels.

Conclusion

The creation of a reliable consumer food scanning application requires a strong and detailed backend database rather than a simple list of E-numbers. An advanced taxonomy must connect E-numbers to their chemical categories, technological functions, and health risk profiles, while also reflecting international regulatory differences across regions such as the EU and ANZ. These variations arise from differing dietary exposures, scientific assessments, and approval processes, which the system must be able to accommodate. Analysis of high-risk additives such as sulphites, azo dyes, benzoates, and MSG shows that health concerns go beyond standard allergies, extending to respiratory sensitivities, behavioural effects, and non-immunological reactions. Unique cases, such as lecithin, highlight the importance of recognising risks linked to source materials rather than the additive itself. Overall, the proposed data model offers a technical foundation for a system that is accurate, adaptable, and capable of providing consumers with precise, context-based information to make informed dietary decisions and manage potential health risks.